

Numerical Methods

By

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Chapter (1): Linear System of Equation

In this chapter we present the solution of n linear simultaneous algebraic equations in n unknowns, we will present three direct methods and two indirect (iterative) methods.

Direct Methods:

1. Doolittle's Method
2. Crout's Method
3. Thomas Algorithm for Tridiagonal System

Indirect or Iterative Methods:

1. Jacobi's Iteration Method
2. Gauss-Seidal Iteration Method

Contents of the Lecture:

Direct Methods:

1. Doolittle's Method
2. Thomas Algorithm for Tridiagonal System

A system of algebraic equations has the form

$$\begin{array}{ccccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & a_{13}x_3 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & a_{23}x_3 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ a_{31}x_1 & + & a_{32}x_2 & + & a_{33}x_3 & + & \cdots & + & a_{3n}x_n & = & b_3 \\ & & & & \vdots & & & & & & \\ a_{n1}x_1 & + & a_{n2}x_2 & + & a_{n3}x_3 & + & \cdots & + & a_{nn}x_n & = & b_n \end{array}$$

Where:

- The coefficients a_{ij} and the constants b_j are known.
- x_i represents the unknowns.

In matrix notation, the equations are written as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

or simply

$$Ax = b$$

- system of linear equations in n unknowns has a unique solution, if $|A| \neq 0$.
- If the coefficient matrix is singular, the equations may have infinite number of solutions, or no solutions at all, depending on the constant vector

- ❑ Direct methods.
- ❑ Iterative methods.

Direct methods:

- Transform the original system into an equivalent system with the same solution but easier to solve.
- The transformation is performed using a finite number of algebraic operations.
- The solution does not involve truncation errors.
- Round-off errors may occur due to floating-point arithmetic.
- The solution is obtained in a finite number of steps.

Iterative methods:

- Start with an initial guess for the solution and repeatedly improve it until a convergence criterion is satisfied.
- The solution contains **truncation errors**.
- Generally less efficient due to the large number of iterations required.
- **Self-correcting**: round-off or arithmetic errors in one iteration may be reduced in subsequent iterations.
- May fail to converge for some systems.
- The initial guess affects only the number of iterations needed for convergence, not the final solution.
- Particularly useful for solving **ill-conditioned system**

LU Decomposition

It is possible to show that any square matrix A can be expressed as a product of a lower triangular matrix L and an upper triangular matrix U .

$$A = LU$$

For instance

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

- The process of computing L and U for a given A is known as LU Decomposition or LU Factorization.
- LU decomposition is not unique, unless certain constraints are placed on L or U . These constraints distinguish one type of decomposition from another.

LU Decomposition

Common Forms of LU Decomposition

1- Doolittle's Method: Constrains are $L_{ii} = 1, i = 1, 2, \dots, n$

2- Crout's decomposition: Constrains are $U_{ii} = 1, i = 1, 2, \dots, n$.

3- Cholesky's Method: Special case for symmetric positive definite matrices $A = LL^t$

After decomposing the matrix A , it is easier to solve the equations

$$Ax = b$$

We can rewrite the equations as

$$LUx = b$$

or denoting $Ux = y$, the above equation becomes

$$Ly = b$$

This equation can be solved for y by forward substitution.

Then $Ux = y$ will yield x by the backward substitution process.

Doolittle's Method

Consider the system of linear equations:

$$\begin{array}{rrcr} a_{11}x_1 & + & a_{12}x_2 & + & a_{13}x_3 = b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & a_{23}x_3 = b_2 \\ a_{31}x_1 & + & a_{32}x_2 & + & a_{33}x_3 = b_3 \end{array}$$

The above system can be written as:

$$Ax = b \quad (1)$$

Where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Doolittle's Method

Let

$$A = LU$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

Equation (1) can be written as

$$LUx = b \quad (2)$$

If we write

$$Ux = V$$

Equation (2) becomes

$$LV = b \quad (3)$$

Equation (3) is equivalent to the system

Doolittle's Method

Equation (3) is equivalent to the system

$$v_1 = b_1$$

$$L_{21}v_1 + v_2 = b_2$$

$$L_{31}v_1 + L_{32}v_2 + v_3 = b_3$$

The above system can be solved to find the values of v_1 , v_2 and v_3 which give us the matrix V .

$$Ux = V$$

then becomes

$$U_{11}x_1 + U_{12}x_2 + U_{13}x_3 = v_1$$

$$U_{22}x_2 + U_{23}x_3 = v_2$$

$$U_{33}x_3 = v_3$$

which can be solved for x_3 , x_2 and x_1 by the backward substitution process.

Doolittle's Method

In order to compute the matrices L and U, we write equation

$$A = LU$$

as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

Multiplying the matrices on the left and equating the corresponding elements of both sides, we obtain

$$U_{11} = a_{11}, U_{12} = a_{12}, U_{13} = a_{13}$$

$$L_{21}U_{11} = a_{21} \rightarrow L_{21} = \frac{a_{21}}{U_{11}} = \frac{a_{21}}{a_{11}}$$

$$L_{31}U_{11} = a_{31} \rightarrow L_{31} = \frac{a_{31}}{U_{11}} = \frac{a_{31}}{a_{11}}$$

Doolittle's Method

$$L_{21}U_{12} + U_{22} = a_{22} \rightarrow U_{22} = a_{22} - L_{21}U_{12} = a_{22} - \frac{a_{21}}{a_{11}}a_{12}$$

$$L_{21}U_{13} + U_{33} = a_{23} \rightarrow U_{23} = a_{23} - L_{21}U_{13} = a_{23} - \frac{a_{21}}{a_{11}}a_{13}$$

$$L_{31}U_{12} + L_{32}U_{22} = a_{32} \rightarrow L_{32} = \frac{1}{U_{22}}(a_{32} - L_{31}U_{12}) = \frac{1}{U_{22}}\left(a_{32} - \frac{a_{31}}{a_{11}}a_{12}\right)$$

$$L_{31}U_{13} + L_{32}U_{23} + U_{33} = a_{33} \quad (4)$$

The value of U_{33} can be computed from Eq. (4)

Example :

Solve the following equations by Doolittle's method:

$$2x + y + 4z = 12$$

$$8x - 3y + 2z = 20$$

$$4x + 11y - z = 33$$

Solution :

We have:

$$A = \begin{bmatrix} 2 & 1 & 4 \\ 8 & -3 & 2 \\ 4 & 11 & -1 \end{bmatrix}, x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, b = \begin{pmatrix} 12 \\ 20 \\ 33 \end{pmatrix}$$

Let:

$$\begin{bmatrix} 2 & 1 & 4 \\ 8 & -3 & 2 \\ 4 & 11 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

Multiplying and equating we get:

$$\begin{bmatrix} 2 & 1 & 4 \\ 8 & -3 & 2 \\ 4 & 11 & -1 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} & L_{21}U_{13} + U_{23} \\ L_{31}U_{11} & L_{31}U_{12} + L_{32}U_{22} & L_{31}U_{13} + L_{32}U_{23} + U_{33} \end{bmatrix}$$

$$U_{11} = 2, U_{12} = 1, U_{13} = 4$$

$$L_{21}U_{11} = 8 \rightarrow L_{21} = 4$$

$$L_{31}U_{11} = 4 \rightarrow L_{31} = 2$$

$$L_{21}U_{12} + U_{22} = -3 \rightarrow 4 + U_{22} = -3 \rightarrow U_{22} = -7$$

$$L_{21}U_{13} + U_{23} = 2 \rightarrow 16 + U_{23} = 2 \rightarrow U_{23} = -14$$

$$L_{31}U_{12} + L_{32}U_{22} = 11 \rightarrow 2 - 7L_{32} = 11 \rightarrow L_{32} = -\frac{9}{7}$$

$$L_{31}U_{13} + L_{32}U_{23} + U_{33} = -1 \rightarrow 8 + 18 + U_{33} = -1 \rightarrow U_{33} = -27$$

We get:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 2 & -\frac{9}{7} & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 4 \\ 0 & -7 & -14 \\ 0 & 0 & -27 \end{bmatrix}$$

and the given system can be written as:

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 2 & -\frac{9}{7} & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 4 \\ 0 & -7 & -14 \\ 0 & 0 & -27 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 12 \\ 20 \\ 33 \end{pmatrix}$$

Writing: $UX = V$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 2 & -\frac{9}{7} & 1 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 12 \\ 20 \\ 33 \end{pmatrix}$$

We get:

$$v_1 = 12$$

$$4v_1 + v_2 = 20 \rightarrow v_2 = -28$$

$$2v_1 - \frac{9}{7}v_2 + v_3 = 33 \rightarrow v_3 = 33 - 24 - 36 = -27$$

Example :

Solve the following equations by Doolittle's method :

$$2x - 6y + 8z = 24$$

$$5x + 4y - 3z = 2$$

$$3x + y + 2z = 16$$

Solution :

We have:

$$A = \begin{bmatrix} 2 & -6 & 8 \\ 5 & 4 & -3 \\ 3 & 1 & 2 \end{bmatrix}, x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, b = \begin{pmatrix} 24 \\ 2 \\ 16 \end{pmatrix}$$

Let:

$$\begin{bmatrix} 2 & -6 & 8 \\ 5 & 4 & -3 \\ 3 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

Multiplying and equating we get:

$$\begin{bmatrix} 2 & -6 & 8 \\ 5 & 4 & -3 \\ 3 & 1 & 2 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} & L_{21}U_{13} + U_{23} \\ L_{31}U_{11} & L_{31}U_{12} + L_{32}U_{22} & L_{31}U_{13} + L_{32}U_{23} + U_{33} \end{bmatrix}$$

$$U_{11} = 2, U_{12} = -6, U_{13} = 8$$

$$L_{21}U_{11} = 5 \rightarrow L_{21} = \frac{5}{2}$$

$$L_{31}U_{11} = 3 \rightarrow L_{31} = \frac{3}{2}$$

$$L_{21}U_{12} + U_{22} = 4 \rightarrow -15 + U_{22} = 4 \rightarrow U_{22} = 19$$

$$L_{21}U_{13} + U_{23} = -3 \rightarrow 20 + U_{23} = -3 \rightarrow U_{23} = -23$$

$$L_{31}U_{12} + L_{32}U_{22} = 1 \rightarrow -9 + 19L_{32} = 1 \rightarrow L_{32} = \frac{10}{19}$$

$$L_{31}U_{13} + L_{32}U_{23} + U_{33} = 2 \rightarrow 12 - \frac{230}{19} + U_{33} = 2 \rightarrow U_{33} = \frac{40}{19}$$

We get:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 2 & 10 & 1 \\ 3 & \frac{10}{19} & 1 \\ \frac{1}{2} & \frac{10}{19} & 1 \end{bmatrix} \begin{bmatrix} 2 & -6 & 8 \\ 0 & 19 & -23 \\ 0 & 0 & \frac{40}{19} \end{bmatrix}$$

and the given system can be written as:

$$\begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 2 & 10 & 1 \\ 3 & \frac{10}{19} & 1 \\ \frac{1}{2} & \frac{10}{19} & 1 \end{bmatrix} \begin{bmatrix} 2 & -6 & 8 \\ 0 & 19 & -23 \\ 0 & 0 & \frac{40}{19} \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 24 \\ 2 \\ 16 \end{pmatrix}$$

Writing: $UX = V$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 2 & 10 & 1 \\ 3 & \frac{10}{19} & 1 \\ \frac{1}{2} & \frac{10}{19} & 1 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 24 \\ 2 \\ 16 \end{pmatrix}$$

We get:

$$v_1 = 24$$

$$\frac{5}{2}v_1 + v_2 = 2 \rightarrow v_2 = 2 - 60 = -58$$

$$\frac{3}{2}v_1 + \frac{10}{19}v_2 + v_3 = 16 \rightarrow v_3 = 16 - 36 + \frac{580}{19} = \frac{200}{19}$$

The solution to the original system is given by:

$$UX = V$$

$$\begin{bmatrix} 2 & -6 & 8 \\ 0 & 19 & -23 \\ 0 & 0 & \frac{40}{19} \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 24 \\ -58 \\ \frac{200}{19} \end{pmatrix}$$

$$2x - 6y + 8z = 24 \rightarrow 2x = 24 + 18 - 40 = 2 \rightarrow \mathbf{x = 1}$$

Example :

Solve the following equations by Doolittle's method :

$$3x + 2y + 7z = 4$$

$$2x + 3y + z = 5$$

$$3x - 4y + z = 7$$

Solution :

Try by yourself

Thomas Algorithm for Tridiagonal System

Consider the system of linear simultaneous algebraic equations given by

$$Ax = b$$

where A is a tridiagonal matrix, $x = [x_1, x_2, \dots, x_n]^t$ and $b = [b_1, b_2, \dots, b_n]^t$.

Hence, we consider a 4×4 tridiagonal system of equations given by

$$\begin{bmatrix} a_{12} & a_{13} & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ 0 & a_{31} & a_{32} & a_{33} \\ 0 & 0 & a_{41} & a_{42} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

The system of equations is solved using Thomas Algorithm which is described in three steps as shown below:

Thomas Algorithm for Tridiagonal System

Step 1: Set $y_1 = a_{12}$ and compute

$$y_i = a_{i2} - \frac{a_{i1}a_{(i-1)3}}{y_{i-1}}, i = 2, 3, \dots, n$$

Step 2: Set $z_1 = \frac{b_1}{a_{12}}$ and compute

$$z_i = \frac{b_i - a_{i1}z_{i-1}}{y_i}, i = 2, 3, \dots, n$$

Step 3:

$$x_i = z_i - \frac{a_{i3}x_{i+1}}{y_i}, i = n-1, n-2, \dots, 1$$

Where:

$$x_n = z_n$$

Example :

Solve the following equations by Thomas Algorithm :

$$\begin{aligned}3x_1 - x_2 &= 5 \\2x_1 - 3x_2 + 2x_3 &= 5 \\x_2 + 2x_3 + 5x_4 &= 10 \\x_3 - x_4 &= 1\end{aligned}$$

Solution :

Here:

$$\begin{bmatrix} 3 & -1 & 0 & 0 \\ 2 & -3 & 2 & 0 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 10 \\ 1 \end{bmatrix}$$

Step 1: Set $y_1 = a_{12} = 3$ and compute

$$y_i = a_{i2} - \frac{a_{i1}a_{(i-1)3}}{y_{i-1}}, i = 2, 3, \dots, n$$

$$i = 2 \quad y_2 = a_{22} - \frac{a_{21}a_{13}}{y_1} = -3 - \frac{(2)(-1)}{3} = \frac{-7}{3}$$

$$i = 3 \quad y_3 = a_{32} - \frac{a_{31}a_{23}}{y_2} = 2 - \frac{(1)(2)}{\left(\frac{-7}{3}\right)} = \frac{20}{7}$$

$$i = 4 \quad y_4 = a_{42} - \frac{a_{41}a_{33}}{y_3} = -1 - \frac{(1)(5)}{\left(\frac{20}{7}\right)} = -\frac{55}{20}$$

Step 2: Set $z_1 = \frac{b_1}{a_{12}} = \frac{5}{3}$ and compute

$$z_i = \frac{b_i - a_{i1} z_{i-1}}{y_i}, i = 2, 3, \dots, n$$

$$i = 2 \quad z_2 = \frac{b_2 - a_{21} z_1}{y_2} = \frac{5 - (2) \left(\frac{5}{3} \right)}{\left(-\frac{7}{3} \right)} = \frac{-5}{7}$$

$$i = 3 \quad z_3 = \frac{b_3 - a_{31} z_2}{y_3} = \frac{10 - (1) \left(-\frac{5}{7} \right)}{\left(\frac{20}{7} \right)} = \frac{75}{20}$$

$$i = 4 \quad z_4 = \frac{b_4 - a_{41} z_3}{y_4} = \frac{1 - (1) \left(\frac{75}{20} \right)}{\left(-\frac{55}{20} \right)} = 1$$

Step 3:

$$x_i = z_i - \frac{a_{i3}x_{i+1}}{y_i}, i = n - 1, n - 2, \dots, 1$$

Where:

$$x_n = z_n$$

$$x_4 = z_4 = 1$$

$$i = 3 \quad x_3 = z_3 - \frac{a_{33}x_4}{y_3} = \frac{75}{20} - \frac{(5)(1)}{\left(\frac{20}{7}\right)} = 2$$

$$i = 2 \quad x_2 = z_2 - \frac{a_{23}x_3}{y_2} = \frac{-5}{7} - \frac{(2)(2)}{\left(\frac{-7}{3}\right)} = 1$$

$$i = 1 \quad x_1 = z_1 - \frac{a_{13}x_2}{y_1} = \frac{5}{3} - \frac{(-1)(1)}{(3)} = 2$$

Example :

Solve the following equations by Thomas Algorithm :

$$\begin{aligned}x_1 + 4x_2 &= 10 \\2x_1 + 10x_2 - 4x_3 &= 7 \\x_2 + 8x_3 - x_4 &= 6 \\x_3 - 6x_4 &= 4\end{aligned}$$

Solution :

Here:

$$\begin{bmatrix} 1 & 4 & 0 & 0 \\ 2 & 10 & -4 & 0 \\ 0 & 1 & 8 & -1 \\ 0 & 0 & 1 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 10 \\ 7 \\ 6 \\ 4 \end{bmatrix}$$

Step 1: Set $y_1 = a_{12} = 1$ and compute

$$y_i = a_{i2} - \frac{a_{i1}a_{(i-1)3}}{y_{i-1}}, i = 2, 3, \dots, n$$

$$i = 2 \quad y_2 = a_{22} - \frac{a_{21}a_{13}}{y_1} = 10 - \frac{(2)(4)}{1} = 2$$

$$i = 3 \quad y_3 = a_{32} - \frac{a_{31}a_{23}}{y_2} = 8 - \frac{(1)(-4)}{2} = 10$$

$$i = 4 \quad y_4 = a_{42} - \frac{a_{41}a_{33}}{y_3} = -6 - \frac{(1)(-1)}{10} = -\frac{59}{10}$$

Step 2: Set $z_1 = \frac{b_1}{a_{12}} = \frac{10}{1} = 10$ and compute

$$z_i = \frac{b_i - a_{i1} z_{i-1}}{y_i}, i = 2, 3, \dots, n$$

$$i = 2 \quad z_2 = \frac{b_2 - a_{21} z_1}{y_2} = \frac{7 - (2)(10)}{(2)} = -\frac{13}{2}$$

$$i = 3 \quad z_3 = \frac{b_3 - a_{31} z_2}{y_3} = \frac{6 - (1)\left(-\frac{13}{2}\right)}{(10)} = \frac{25}{20}$$

$$i = 4 \quad z_4 = \frac{b_4 - a_{41} z_3}{y_4} = \frac{4 - (1)\left(\frac{25}{20}\right)}{\left(-\frac{59}{10}\right)} = -\frac{55}{118}$$

Step 3:

$$x_i = z_i - \frac{a_{i3}x_{i+1}}{y_i}, i = n - 1, n - 2, \dots, 1$$

Where:

$$x_n = z_n$$

$$x_4 = z_4 = -\frac{55}{118} = -0.466$$

$$i = 3 \quad x_3 = z_3 - \frac{a_{33}x_4}{y_3} = \frac{25}{20} - \frac{(-1)\left(-\frac{55}{118}\right)}{(10)} = 1.203$$

$$i = 2 \quad x_2 = z_2 - \frac{a_{23}x_3}{y_2} = -\frac{13}{2} - \frac{(-4)(1.203)}{(2)} = -4.094$$

$$i = 1 \quad x_1 = z_1 - \frac{a_{13}x_2}{y_1} = 10 - \frac{(4)(-4.094)}{(1)} = 26.376$$

Example :

Solve the following equations by Thomas Algorithm :

$$\begin{aligned}2x_1 - x_2 &= 1 \\x_1 + 3x_2 + x_3 &= 3 \\x_2 + x_3 + 2x_4 &= 0 \\2x_3 + 3x_4 &= -1\end{aligned}$$

Solution :

Try by yourself

**THANK YOU
FOR YOUR
ATTENTION**

